

Student Takeaway for Sync Session

<<Week 10: Applications of AI and ML – Images, Video & Speech >>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

Session title	Applications of AI and ML – Images, Video & Speech	
Session duration	2 hours	
Session type	Lectures: Understanding visual and audio data, AI pipelines, real-world use cases, and current challenges. Demo & Case: based Discussions: Failures, limitations, ethics, and bias in vision/speech AI.	
Scope	This session introduces learners to how AI "sees" and "hears" the world through computer vision and audio processing. Topics include: - Core AI pipelines for image, video, and audio. - Industry applications across security, healthcare, retail, and accessibility. - Limitations, failures, ethical issues, and bias. - Future trends in multimodal and generative AI.	
Learning objectives	Objective	Core capability
	Understand how machines process visual and audio data	Conceptual understanding of computer vision and speech processing
	Identify key real-world applications	Real-world relevance of AI/ML technologies
	Recognise system failures and limitations	Critical thinking about model behaviour
	Explore ethical and societal implications	Awareness of bias, privacy, and fairness
Software/Tools	Vision: CNNs, image classification, object detection Speech: Spectrograms, RNNs/Transformers, Whisper, TTS Demos: ChatTube, adversarial image examples Datasets: Audio/image samples, case study clips	

Pro tips for success:

Ask Bold Questions: Clarify how adversarial attacks fool AI (e.g., stop sign stickers).

Be Hands-On: Try Chattube.io to query a YouTube video.

Collaborate: Debate AI surveillance trade-offs in breakout rooms.

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1-3	Session Agenda	Overview of vision/speech AI and multimodal trends.	Focus on real-world impact.
4	Async Recap	Recap of AI in images (classification, captions), video (action tracking), and audio (speech/text). Reflect on what helped you most.	Labels, captions, objects
5	How AI "Sees" and "Hears"	Pixels (RGB grids) vs. audio (spectrograms).	CNNs for images, transformers for audio.
6	AI Pipeline	Input → preprocessing → model → output.	Demo: ResNet50 image classification.
7	Real AI Mistakes	Misclassified objects, voice assistant errors.	Discuss edge cases (accents, rare objects).
8	Why AI Fails	Data bias, adversarial attacks, environmental noise.	Case: Gender Shades study (34.7% error rate).
9	Business Applications	Retail (cashierless stores), healthcare (MRI analysis).	Highlight cost-value trade-offs.
10	Ethics & Bias	Surveillance risks, representational bias.	Debate: AI cameras in cities.
11-12	Case Studies	Adversarial stop signs, biased voice recognition.	MIT study on facial recognition.
13	Where Is Vision & Speech AI Used?	A look at real-world AI applications in security, healthcare, social media, retail, accessibility, and autonomous vehicles	Face ID, surveillance Healthcare: Scan analysis, monitoring
14	Future Trends	Multimodal AI (GPT-4o), generative media (deepfakes).	Risks: Misinformation, voice cloning.
15	Risks and Societal Impact	Key risks of Vision/Speech AI: deepfakes, job loss, surveillance, and accountability issues.	Fake media, scams AI replaces moderators, transcribers
16	Group Scenario / Reflection	Debate on AI cameras in cities—weighing safety benefits vs. privacy/freedom risks.	Traffic control, crime reduction, AMBER alerts
17-20	Chat With YouTube Video	Demo: Use Chattube.io to query video content.	Try asking about key points.

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21	Q&A	Open discussion on tools and ethics.	Clarify doubts about demos.

Post session

Reflection challenge	Which AI limitation (bias, adversarial attacks) concerns you most? Why?
Explore more	• Read: Gender Shades study. • Try: Fine-tune Whisper on custom audio. • Experiment: Test YOLOv8 on personal images.
Get inspired	AI vision powers life-saving tools like fall detection for the elderly—your skills could expand accessibility!
The journey ahead	Next week: Advanced Model Interpretability—learn to debug and explain AI decisions.